

Jacob Horne

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EDUCATION

University of California, Irvine

June 2028

Bachelor of Science in Computer Science; Campuswide Honors Collegium

GPA: 3.92

Coursework: Object-Oriented Programming (C/C++), Data Structures & Algorithms, Databases, Computer Architecture

EXPERIENCE

Sandia National Laboratories

Fall 2026

Incoming Software Engineering Intern — AI R&D Toolkits

Livermore, CA

Capital One

June 2026 – Present

Software Engineering Intern — ML Recommender Infrastructure

New York, NY

- Replaced a network API call with an in-process SDK to reduce latency by **25%** for **60M+ customers monthly**
- Built Databricks/PySpark sampling pipeline processing **800M+** historical customer-message outcomes, enabling new candidate recommender models to be tested against production behavior over broader date ranges **28× faster**
- Automated sampling and upstream jobs in Apache Airflow with **18** schema, distribution, and multiplier quality checks

UCI Digital Learning Lab

Dec 2025 – Present

Research Assistant — LLM Systems and Evaluation

Irvine, CA

- Built Python-based multi-agent essay-grading pipelines with LangGraph, implementing critic, advocate, judge agents
- Analyzed LLM output datasets using NumPy, Pandas, and Matplotlib for token log-probs, behavior and confidence
- Co-authored [paper](#) accepted to **ICLR 2026** LLM Workshop on multi-agent evaluation and token-level LLM confidence
- Investigating multi-turn text-to-SQL confidence using token log-probabilities through query execution and repair

Calit2

Apr 2026 – Present

AI/ML Undergraduate Researcher — Physics-Informed Machine Learning

Irvine, CA

- Built and optimized a physics-informed PyTorch/GRU time-series pipeline with heat-transfer residual losses and initial-condition constraints to replace hours-long FEA simulations with fast cylindrical wall temperature predictions
- Accelerated model training with MPS/GPU support, gradient clipping, bounded parameter discovery, and multi-seed benchmarking to compare PINN, LSTM, and GRU architectures across timestep resolutions and thermal scenarios
- Achieved **R-squared = 0.99** with **12,000× faster inference** than FEA, enabling scalable heat-transfer modeling

CareTech, UCI

Oct 2025 – Jun 2026

Software Developer — Computer Vision

Irvine, CA

- Built a Next.js/FastAPI food-scanning app for real-time model inference and automated vitamin and nutrient tracking
- Integrated PyTorch inference with backend APIs and nutrition data, converting food predictions into live intake totals
- Processed and augmented **28K+ food images** across **270+ categories**, training PyTorch models to **81% accuracy**

Commit the Change, UCI

Oct 2025 – Jun 2026

Full-Stack Developer — Backend and Platform

Irvine, CA

- Built Node.js/Express APIs processing **5,000+ weekly requests** for a scheduling platform serving **18,000+ patients**
- Reduced booking conflicts and manual scheduling workload by **30%** with real-time appointment and quota tracking

PROJECTS

Agonus (Blockchain, UCI) | LangChain, FastAPI, Next.js, Solidity, PostgreSQL

Oct 2025 – Apr 2026

- Built a decentralized AI trading platform where autonomous agents compete in crypto tournaments with on-chain betting, real-time performance tracking with leaderboard, tokenized reward incentives, and a live tournament dashboard UI
- Architected LangChain ReAct agents (GPT-4o-mini) with custom tools for market data, trade execution, planning, and memory; designed Celery/Redis orchestration with scheduling, locking, crash recovery to manage **10+** concurrent agents

Tata Consultancy Services Capstone | FastAPI, PostgreSQL, pgvector, OpenAI, DeBERTa

Jan 2026 – Jun 2026

- Co-developed a React/FastAPI banking chatbot with DeBERTa ethics checks for safe, compliant customer guidance
- Owned an end-to-end document RAG pipeline with PDF/OCR ingestion, document classification, figure extraction, OpenAI embeddings, PostgreSQL/pgvector retrieval, and banking-metric analysis to ground responses in customer data

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C/C++, SQL, Java, R, MIPS Assembly

ML/Data: PyTorch, scikit-learn, TensorFlow, PySpark, OpenCV, LangChain, LangGraph, NumPy, Pandas, Matplotlib

Platforms/Tools: Databricks, Airflow, PostgreSQL, Redis, Celery, FastAPI, Docker, Git, AWS (ECS, Fargate, S3, CloudWatch)